Redactor: Select-Once Object Redaction for Video

CIS 5810 Final Project Proposal • Group 34 • Option 1

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1. Goals and audience

Anyone who posts footage of public places (journalists, researchers, people filming a street or an event) eventually needs to hide a bystander, a face, or a license plate. Right now that means keyframing a blur by hand, shot by shot. We want a tool where you say once what to hide and it stays hidden for the rest of the clip. For now you do that by outlining the object on the first frame, but we are building the pipeline so the request could also come from a detector or a text description. We care most about leaks (the object should never be visible, even for a frame), and then about keeping the mask tight so we don't blur the rest of the scene.

2. Pipeline design

We organized the pipeline around a single shared data type that we call the target prompt: a rough region on some frame (a box or a coarse mask) plus an instance id. The stages before the prompt decide what to hide. The stages after it work out where that thing is in each frame and hide it (Figure 1).

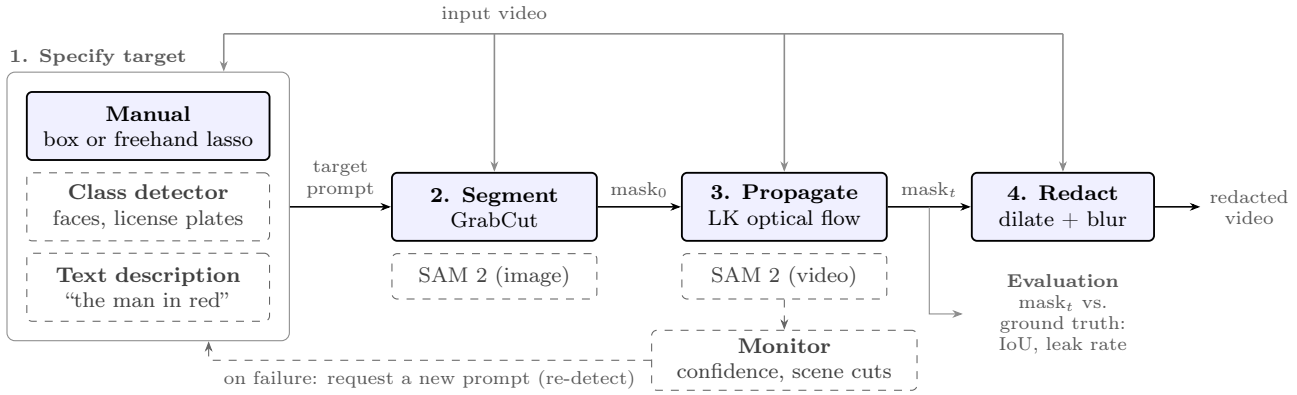


Figure 1: Our pipeline. Every target source produces the same kind of prompt. Solid boxes are built, dashed boxes are planned.

1. **Specify target:** Right now the user draws on the first frame and checks a preview of the segmentation before anything runs. We want to add two more sources that plug into the same interface. A class detector would propose every face or license plate with no user input. A text description would go through an open-vocabulary detector such as Grounding DINO [7], which returns boxes for a free-form phrase. These two behave differently from manual selection: they return several instances at once and they can fire on any frame, so something that enters the clip late still gets a prompt. The later stages will have to handle both.
2. **Segment:** GrabCut [1] turns the rough prompt into a pixel mask by fitting color models to the foreground and background.
3. **Propagate:** We find Shi–Tomasi corners [2] inside the current mask, follow them into the next frame with pyramidal Lucas–Kanade flow [3], and shift the mask by their median displacement.
4. **Redact:** We dilate the mask with a 9×9 kernel as a safety margin and replace that region with a 41×41 Gaussian blur.

Why this design: We wanted to be able to swap out one stage and measure it on its own, which works because the benchmark reads the mask between stages 3 and 4. We started with classical methods because they need only a CPU and OpenCV, and when they fail you can see why. That gives us something concrete to compare SAM 2 against in stages 2 and 3. The monitor is the piece we are missing today: when confidence drops or the scene cuts, it should ask stage 1 for a new prompt.

3. Baseline results

Baseline: The user drags a bounding box, GrabCut is initialized from that rectangle, and propagation is the translation-only tracker described above.

Evaluation: We don't have annotated real footage yet, so we wrote a synthetic benchmark (`scripts/evaluate.py`). It renders a textured, person-shaped object over a cluttered background, which gives us exact ground-truth masks. There are four 90-frame clips, each testing one thing: plain translation, scale change (the object shrinks to 45% as it walks away), occlusion (it passes completely behind a pillar), and a scene cut at frame 45. We simulate the user's prompt from the ground truth with 5, 9 and 13 pixels of slack and average the three runs. We report mask IoU against the visible object and a leak rate, which is the fraction of visible object pixels still unblurred after the safety dilation. For a privacy tool the leak rate is the number we care about most.

	Frame 0	Translation		Scale		Occlusion		Scene cut	
Method	IoU	IoU	Leak	IoU	Leak	IoU	Leak	IoU	Leak
Baseline: box + GrabCut + LK translation	0.87	0.85	0%	0.41	22%	0.37	49%	0.43	50%
Iteration 1: lasso + seeded GrabCut	0.84	0.84	0%	0.42	18%	0.36	49%	0.42	50%
Iteration 1 + occlusion gate (prototype)	0.84	0.84	0%	0.42	17%	0.36	49%	0.42	50%

Table 1: Mean mask IoU (higher is better) and mean leak rate (lower is better) over each 90-frame synthetic clip.

The baseline does fine when the object only translates (IoU 0.85, no leak) and breaks in the other three clips (Figure 2). When the object shrinks, the mask stays its original size and slides off, hitting IoU 0 by frame 80. In the occlusion clip the mask gets stuck at the edge of the pillar and never catches up, so IoU is 0 for every frame after the object comes back out. After the scene cut the mask just stays where it was, and the object is fully exposed for the rest of the clip.

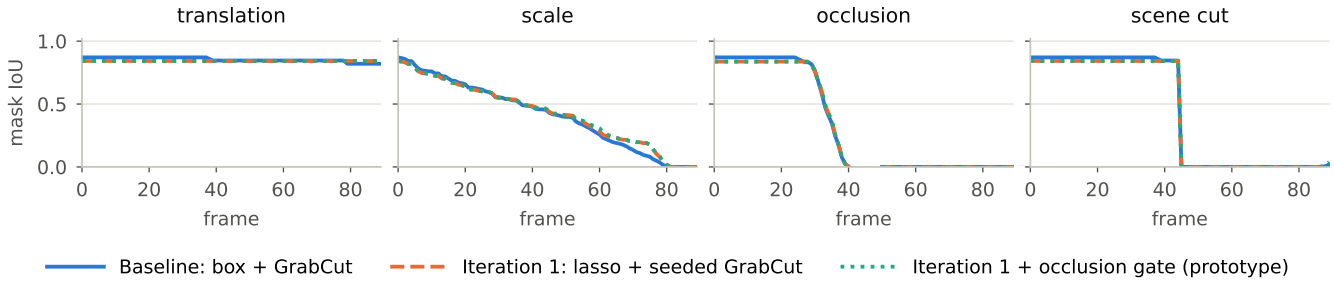


Figure 2: Mask IoU per frame on the synthetic clips. The gap under occlusion is where the object is fully hidden.

4. Iteration 1 results

Change: A box is a poor fit for a pedestrian in a crowd, because a rectangle around one person also picks up parts of the people next to them. So we replaced the box with a freehand lasso and seeded GrabCut from it. Pixels deep inside the outline (more than 35% of the maximum distance from its edge) are marked definite foreground, the rest of the outline is probable foreground, a thin band just outside is probable background, and everything else is definite background. Propagation and redaction did not change.

Synthetic result: On the benchmark the lasso did not help (Table 1). First-frame IoU is 0.84 versus 0.87 for the box, and the tracking numbers are within 0.01 IoU of the baseline in every clip. We think this is because the synthetic object stands alone, so either prompt gives GrabCut enough to work with. The benchmark can't show what the lasso is for, and we will need crowded real frames with annotated masks to measure that. It also tells us that segmentation is not what is holding us back. Every failure in Table 1 comes from propagation.

Real footage: Figure 3 shows Iteration 1 on a 5-second clip of a crowded crosswalk (2320×1080, 150 frames). The selection on frame 0 looks right. By frame 75 the blur is lagging behind the subject and covering part of the person next to her. By frame 148 she has walked away and gotten smaller, the mask has not, and it is now blurring bystanders in the foreground while she is fully visible. This is the same scale failure we saw on the benchmark, plus drift onto nearby people. Video: https://drive.google.com/file/d/1xmE9le1VtFV0ecj26mFJQgkK42axJxDS/view?usp=sharing.

Occlusion gate (prototyped, then removed): We also tried to catch bad motion estimates with a forward-backward check [4]: track the points forward, track them back, and keep the ones that land within 1.5 pixels of where they started. If fewer than three survive, or the jump is over 20 pixels, we freeze the mask. It made no difference (Table 1). On the occlusion clip the gate never fired, because the points that are left as the object disappears still pass the check, and the mask gets stuck at the pillar just like before. On the scene-cut clip it fired on every frame after the cut, but a frozen mask does not find the object again, so we took it out. Noticing a failure does not help unless we can also re-detect the object.



Figure 3: Iteration 1 on real footage (frames 0, 75, 148). Top: input. Bottom: output. Subject: the woman in the white skirt.

5. Plan for improvements

Each iteration changes one stage, and we score it on the same benchmark. Items 2–4 go after the scale, occlusion and scene-cut clips in that order.

1. **Evaluation on real video:** Add sequences from DAVIS 2017 [5], which comes with per-frame object masks and has sequences with occlusion and scale change. We will also annotate frames from our own crosswalk footage.
2. **Scale-aware propagation:** Fit a similarity transform to the tracked points with RANSAC instead of using the median translation, and re-run GrabCut on the propagated mask every few frames to keep it from drifting.
3. **Occlusion recovery:** Keep an appearance model of the object from frame 0 (a color histogram first, learned features later). When confidence drops, hold the mask on its predicted path and search for the object.
4. **Scene cuts:** Detect cuts from the frame-to-frame histogram difference, then trigger the same re-detection, or stop redacting if the object is gone.
5. **SAM 2:** Replace stages 2 and 3 with SAM 2 [6], which propagates a prompted mask through a video using a memory of past frames and is built to handle occlusion and reappearance. We will compare it with our classical pipeline on accuracy and speed.
6. **New target sources:** Add the class detector (faces, license plates) and then the text-description source [7] to stage 1. This also means extending propagation to several instances and to prompts that arrive mid-clip.

6. Timeline, milestones, and member duties

Dates	Milestone	Lead
10/12–10/25	Real-video benchmark (DAVIS + annotated crosswalk frames); first TA meeting	Dhruv
10/19–11/1	Scale-aware propagation with periodic re-segmentation	Kevin
10/26–11/8	SAM 2 integrated as an alternative segment/propagate backend	AJ
11/2–11/15	Occlusion recovery and scene-cut handling	Kevin
11/20	Midterm report: three iterations compared on the benchmark	All
11/21–12/3	Detector and text-description target sources; UI polish	AJ, Dhruv
12/6	Final Project Day demo	All
12/10	Final report, demo video, code	All

Duties: Kevin is taking propagation (tracking, occlusion, cuts). AJ is taking target selection and segmentation, including SAM 2. Dhruv is taking the evaluation data and metrics, plus the detector and text sources.

TA request: None.

Generative AI use: Claude Code (Anthropic) has been used for helping develop evaluation scripts and graphs, and for drafting ideas of where to take this project for this proposal

References

[1] C. Rother, V. Kolmogorov, A. Blake. GrabCut: Interactive foreground extraction using iterated graph cuts. *SIGGRAPH*, 2004.

[2] J. Shi, C. Tomasi. Good features to track. *CVPR*, 1994.

[3] B. Lucas, T. Kanade. An iterative image registration technique with an application to stereo vision. *IJCAI*, 1981.

[4] Z. Kalal, K. Mikolajczyk, J. Matas. Forward-backward error: Automatic detection of tracking failures. *ICPR*, 2010.

[5] J. Pont-Tuset et al. The 2017 DAVIS challenge on video object segmentation. *arXiv:1704.00675*, 2017.

[6] N. Ravi et al. SAM 2: Segment anything in images and videos. *arXiv:2408.00714*, 2024.

[7] S. Liu et al. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. *arXiv:2303.05499*, 2023.